A Project Report on

Real-Time Barcode-Enabled Inventory Tracking

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING

By

Akshat Mukkawar : 24106190

Atharva Mayekar : 24106065

Rohit Nahar : 24106055

Tejas Maske : 24106134

Under the Guidance of

Prof. Viki Patil

Department of Computer Science & Engineering

(Artificial Intelligence & Machine Learning)

A.P. SHAH INSTITUTE OF TECHNOLOGY, THANE

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A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

CERTIFICATE

This is to certify that the project entitled "Real-Time Barcode-Enabled Inventory Tracking" is a bonafide work of **Akshat Mukkawar(24106190), Atharva Mayekar(24106065), Rohit Nahar(24106055), Tejas Maske(24106134)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**).

Prof. Viki Patil
Guide

Prof. Vijaya Bharathi J
Project Co-ordinator

Dr. Jaya Gupta
Head of Department

Dr. Uttam Kolekar
Principal

A. P. SHAH INSTITUTE OF TECHNOLOGY, THANE

Project Report Approval for S.E.

This project report entitled ***Real-Time Barcode-Enabled Inventory Tracking*** by ***Akshat Mukkawar(24106190), Atharva Mayekar(24106065), Rohit Nahar(24106055), Tejas Maske(24106134)*** is approved for the degree of ***Bachelor of Engineering*** in ***Computer Science & Engineering (Artificial Intelligence & Machine Learning)***, ***2025-26***.

Examiner Name	Signature
1._____	_____
2._____	_____

Date:

Place:

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(Signature)
Akshat Mukkawar - 24106190

(Signature)
Atharva Mayekar - 24106065

(Signature)
Rohit Nahar - 24106055

(Signature)
Tejas Maske - 24106134

Date:

Abstract

The rapid growth of small retail businesses has increased the need for efficient and accurate inventory management systems. Traditional methods of inventory management, such as manual record-keeping and basic digital tools, are often time-consuming, error-prone, and lack real-time updates. This project proposes a barcode-based inventory management system that automates stock tracking using a webcam and image processing techniques. The system utilizes barcode scanning to identify products and update inventory records in real time, thereby reducing manual effort and minimizing human errors.

The proposed system is developed using Python and Flask for backend processing, OpenCV and pyzbar for barcode detection, and MySQL for database management. It provides a user-friendly interface and eliminates the need for dedicated barcode scanning hardware, making it cost-effective and accessible for small businesses. The system demonstrates improved accuracy, faster processing time, and enhanced efficiency compared to traditional methods.

This project aligns with the Sustainable Development Goal 9 by promoting technological innovation and supporting the development of efficient infrastructure for small-scale industries. Additionally, it contributes to Sustainable Development Goal 8 by enhancing productivity and operational efficiency in retail businesses.

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ABBREVAION

API	*Application Programming Interface*
BARCODE	*Barcode Technology*
CV	*Computer Vision*
DB	*Database*
FLASK	*Python Web Framework*
GUI	*Graphical User Interface*
HTML	*HyperText Markup Language*
HTTP	*HyperText Transfer Protocol*
IDE	*Integrated Development Environment*
JS	*JavaScript*
JSON	*JavaScript Object Notation*
OS	*Operating System*
OPENCV	*Open Source Computer Vision Library*
RAM	*Random Access Memory*
SKU	*Stock Keeping Unit*
SQL	*Structured Query Language*
MYSQL	*Lightweight Database Engine*
UI	*User Interface*
URL	*Uniform Resource Locator*

Chapter 1

Introduction

1.1 Introduction

Inventory management plays a vital role in the smooth functioning of any business, especially in the retail sector where the availability and tracking of products directly impact sales and customer satisfaction. In small retail businesses, inventory is often managed using traditional methods such as handwritten registers or simple spreadsheet tools. While these methods are easy to implement, they are inefficient, time-consuming, and highly prone to human errors. Mistakes in data entry, duplication of records, and lack of real-time updates can lead to inaccurate stock information, resulting in overstocking, understocking, and financial losses.

With the advancement of technology, various digital inventory management systems have been developed to overcome these challenges. However, many of these systems are either too complex or too expensive for small-scale businesses. They often require specialized hardware such as barcode scanners and involve installation and maintenance costs. Additionally, these systems may require technical expertise, making them less accessible to users who are not familiar with advanced technologies.

To address these issues, this project proposes a barcode-based inventory management system that uses a webcam for scanning and image processing techniques for barcode detection. The system is designed to be simple, cost-effective, and efficient, making it suitable for small retail businesses. By eliminating the need for dedicated hardware and using open-source technologies, the system reduces implementation costs while maintaining high performance.

The proposed system integrates barcode scanning, database management, and a web-based interface to automate inventory operations. Each product is assigned a unique barcode that acts as an identifier. When a barcode is scanned, the system retrieves product details from the database and updates stock levels in real time. This automation reduces manual effort, minimizes errors, and improves overall efficiency.

The system is developed using modern tools and technologies such as Python and Flask for backend processing, OpenCV and pyzbar for barcode detection, and MySQL for database management. The user interface is designed to be simple and intuitive, allowing users to perform operations such as adding products, updating stock, and viewing inventory records easily. Overall, the system aims to provide a practical and accessible solution for inventory management in small businesses.

1.2 Background and Motivation

Inventory management has been a fundamental aspect of business operations for many years. In small businesses, inventory is often managed manually due to limited resources and lack of access to advanced technologies. Although manual systems are easy to use, they become inefficient as the number of products increases. Errors in maintaining stock records can lead to serious issues such as loss of sales, customer dissatisfaction, and poor decision-making.

The motivation behind this project is to develop a system that overcomes the limitations of traditional and existing digital systems while remaining affordable and easy to use. The increasing availability of low-cost computing devices and open-source software provides an opportunity to develop efficient solutions without significant investment. By leveraging these technologies, it is possible to create a system that improves accuracy, reduces manual effort, and provides real-time updates.

Another important motivation is to eliminate the dependency on specialized hardware such as barcode scanners. Many small businesses cannot afford such equipment, which limits thei

ability to adopt automated systems. By using a webcam for barcode scanning, the proposed system offers a cost-effective alternative without compromising functionality. The project also focuses on improving usability by designing a simple and user-friendly interface that does not require technical expertise.

1.3 System Overview

The proposed inventory management system is designed to automate stock tracking using barcode technology and image processing. The system consists of three main components: the frontend interface, the backend processing unit, and the database. The frontend provides an interface for users to interact with the system, while the backend handles data processing and communication between components. The database stores product details and transaction records The system workflow begins when a user scans a barcode using a webcam. The captured image is processed using OpenCV, and the barcode is decoded using the pyzbar library. The decoded information is sent to the backend, where the system checks the database for the corresponding product. If the product exists, its details are retrieved and the stock is updated. If the product is not found, the user is prompted to add a new product.

This approach ensures real-time updates and accurate inventory tracking. It also maintains transaction history, which helps in analyzing stock movement and business performance. The system is designed to be efficient, reliable, and easy to use.

Beyond basic tracking, the system has been enhanced into a comprehensive Retail Management solution. It now features a robust Point of Sale (POS) billing system and an Analytics Engine. This allows for multi-product cart management, real-time financial calculations including GST and discounts, and high-level business intelligence. By capturing "Buy, Sell, and Return" movements, the system provides actionable insights into revenue trends and product performance.

1.3.1 Scope of the Project

The scope of this project is focused on developing a basic yet efficient inventory management system for small retail businesses. The system handles essential operations such as product addition, stock updating, and inventory tracking. It uses barcode scanning to automate these processes and reduce manual effort.

The system is limited to a single-user environment and uses a local database for storing data. Advanced features such as cloud integration, multi-user access, and predictive analytics are not included in the current version. However, the system is designed to be scalable, allowing future enhancements to be added easily. The project also includes testing the system under different conditions to evaluate its performance and reliability.

1.4 Objectives of the Proposed System

The primary objective of the proposed system is to develop an efficient and reliable inventory management solution that addresses the limitations of existing systems. The system aims to provide real-time stock tracking, reduce human errors, and improve overall efficiency. It focuses on minimizing manual effort by automating inventory operations using barcode scanning.

Another important objective is to create a cost-effective solution that does not require specialized hardware. By using a webcam for barcode detection, the system reduces implementation costs and increases accessibility for small businesses. The project also aims to design a user-friendly interface that allows users to operate the system without requiring technical expertise.

Additionally, the system aims to maintain accurate transaction records and provide quick access to inventory data. This helps in better decision-making and improves overall business performance. The proposed system also ensures scalability, allowing future enhancements to be integrated as needed.

1.5 Advantages of the Proposed System

- Reduction of Human Errors: The system automates data entry, which ensures accurate stock updates in real time.
- Improved Efficiency: It reduces the time required for inventory operations and allows users to manage stock more effectively.
- Cost-Effectiveness: By using a webcam instead of dedicated barcode scanners, it provides an affordable solution for small businesses.
- Ease of Use: The interface is designed to be accessible to a wide range of users without requiring advanced technical knowledge.
- Transaction Tracking: The system maintains proper records, which helps in tracking inventory movement and analyzing overall business performance.
- Flexibility and Scalability: The architecture is designed to be easily extended with new features and enhancements in the future.
- Practicality: Overall, it provides a highly efficient and practical solution for modern inventory management.

Chapter 2

Literature Survey/ Existing System

2.1 Literature Survey

The study of inventory management systems has gained significant attention in recent years due to the increasing need for efficient stock tracking and management in retail and industrial environments. Various research works have been carried out to improve inventory accuracy, reduce manual effort, and enhance real-time tracking capabilities. This section reviews existing literature related to inventory management systems, barcode technology, image processing, and database integration, highlighting their contributions and limitations.

2.2 Traditional and Digital Inventory Systems

Early inventory management systems were primarily manual, relying on handwritten records and basic spreadsheets. While these methods were simple to implement, they lacked accuracy and efficiency. According to S. Smith [1], manual inventory systems are highly prone to human errors, including incorrect data entry and duplication of records, which can lead to inaccurate stock management. These limitations make manual systems unsuitable for businesses dealing with a large number of products.

With the advancement of technology, digital inventory systems were introduced to overcome these challenges. J. Lee [2] proposed a web-based inventory management system that allows real-time tracking and remote accessibility. The study highlights that digital systems significantly improve accuracy and efficiency compared to manual methods. However, such systems often require stable internet connectivity and technical expertise, which can be a limitation for small-scale businesses.

Further research by M. Brown [5] emphasizes the importance of database management systems in inventory applications. The study demonstrates how structured databases improve data storage, retrieval speed, and consistency. However, it also points out that improper database design can lead to inefficiencies and data inconsistencies.

2.3 Barcode Technology in Inventory Management

Barcode technology has become a widely used solution for automating inventory management processes. It enables quick and accurate identification of products, reducing the need for manual data entry. K. Sharma [4] discusses the implementation of barcode and QR code-based inventory systems, highlighting their ability to improve operational efficiency and reduce human errors.

In another study, P. Kumar [8] explores inventory optimization techniques using barcode systems. The research shows that barcode-based systems significantly reduce processing time and improve accuracy. However, the study also identifies limitations such as the need for dedicated hardware and initial setup costs.

2.3.1 Image Processing and Barcode Detection

Recent advancements in image processing have enabled the use of cameras for barcode detection, eliminating the need for dedicated scanning devices. R. Gupta [3] presents a method for barcode detection using OpenCV and image processing techniques. The study demonstrates that camera-based barcode scanning can achieve high accuracy under proper conditions.

The use of libraries such as OpenCV and pyzbar has further simplified the implementation of such systems. These tools allow real-time detection and decoding of barcodes from images captured using a webcam. However, the performance of these systems can be affected by factors such as lighting conditions, camera quality, and image resolution.

This approach provides a cost-effective alternative to traditional barcode scanners, making it particularly suitable for small businesses. It also opens up new possibilities for integrating image processing with inventory management systems.

2.4 Web-Based Frameworks and System Development

Modern inventory systems often rely on web-based frameworks for implementation. D. Wilson [7] discusses the development of web applications using the Flask framework, highlighting its simplicity and flexibility. Flask allows developers to create lightweight and scalable applications, making it an ideal choice for small to medium-scale projects.

Web-based systems provide several advantages, including remote accessibility, centralized data management, and ease of maintenance. However, they also introduce challenges such as security concerns and dependency on internet connectivity. These factors must be considered while designing inventory management systems.

The integration of frontend technologies such as HTML, CSS, and JavaScript with backend frameworks enables the development of user-friendly interfaces. This improves user experience and makes the system more accessible to non-technical users.

2.5 Research Gap and Need for Proposed System

The literature survey indicates that while existing systems offer various advantages, they also have significant limitations. Manual systems lack accuracy and efficiency, while digital systems are often complex and expensive. Barcode-based systems improve performance but require dedicated hardware, increasing the cost of implementation.

Image processing-based systems provide a promising alternative by using cameras for barcode detection. However, their performance depends on environmental conditions, which can affect reliability. Additionally, many existing systems do not focus on simplicity and usability, making them difficult for small business owners to adopt.

There is a clear need for a system that combines the advantages of existing approaches while minimizing their limitations. The proposed system aims to address this gap by providing a cost-effective, easy-to-use, and efficient inventory management solution. By using a webcam for barcode scanning and integrating it with a lightweight web-based framework, the system eliminates the need for specialized hardware and reduces implementation complexity.

This approach ensures real-time updates, improved accuracy, and better usability, making it suitable for small retail businesses. The proposed system bridges the gap between traditional

Sr	Author	Method Used	Key Findings	Limitations
1	Smith (2020)	Manual System	Simple to implement and requires minimal technical knowledge	Highly prone to human errors, time-consuming, and inefficient for large datasets
2	Lee (2021)	Web-Based System	Enables real-time updates, remote accessibility, and improved coordination	Requires a stable internet connection and basic technical infrastructure
3	Gupta (2019)	OpenCV Detection	Provides a low-cost and automated scanning solution with minimal hardware	Performance highly dependent on lighting conditions and camera quality
4	Sharma (2022)	Barcode System	Significantly improves accuracy, speed, and reduces manual workload	Requires dedicated barcode hardware and initial setup cost
5	Wilson (2020)	Flask-Based System	Lightweight, scalable, and easy to deploy for small to medium applications	Limited advanced features and may require integration with other technologies
6	Patel (2021)	RFID-Based System	Enables fast and contactless tracking with high efficiency	High implementation cost and requires specialized RFID hardware
7	Kumar (2023)	AI-Based Detection System	Provides intelligent automation with high accuracy and adaptability	Requires large datasets and higher computational resources

Table 1: Summary of Literature Survey

Chapter 3

Limitation of Existing system

3.1 Limitation of Existing System

Inventory management systems currently in use, whether manual or digital, exhibit several limitations that affect their efficiency, accuracy, and usability. Traditional manual systems rely heavily on handwritten records or basic spreadsheet tools, which are highly prone to human errors such as incorrect data entry, duplication, and miscalculations. These systems are also time-consuming and do not provide real-time updates, making it difficult for business owners to track inventory effectively. As a result, issues such as overstocking, understocking, and loss of inventory are commonly observed.

Digital inventory systems have improved the process by introducing automation; however, they still depend on manual input for updating stock data. This increases the chances of errors and reduces efficiency. Moreover, many of these systems are complex and require technical knowledge, making them less suitable for small-scale businesses.

3.2 Limitations of Manual Systems

Manual inventory systems are simple but inefficient. They require constant human involvement for recording and updating stock details. This leads to a high probability of errors and inconsistencies in data. Additionally, retrieving information from manual records is time-consuming and difficult, especially when dealing with a large number of products. The absence of real-time updates further reduces their effectiveness in dynamic business environments.

3.3 Limitations of Existing Digital Systems

Although digital systems offer better performance than manual methods, they come with their own set of challenges. Many digital inventory systems are expensive to implement and maintain. They often require installation of software, regular updates, and technical support. Furthermore, these systems may not be user-friendly for individuals with limited technical knowledge, which restricts their usability in small businesses.

3.4 Limitations of Barcode-Based Systems

Barcode-based inventory systems improve accuracy and speed by automating product identification. However, these systems typically require dedicated barcode scanners and specialized hardware. This increases the overall cost of implementation. In addition, hardware failures or maintenance issues can disrupt system functionality, leading to operational delays.

3.5 Limitations of Image Processing Systems

Image processing-based systems, which use cameras for barcode detection, provide a cost-effective alternative to traditional barcode scanners. However, their performance is influenced by external factors such as lighting conditions, camera resolution, and image quality. Poor lighting or low-quality images can result in incorrect barcode detection, affecting system reliability.

3.6 Research Gap

The analysis of existing systems highlights a clear gap in the availability of a solution that is both efficient and affordable for small businesses. While advanced systems offer automation and real-time tracking, they are often expensive and require technical expertise. On the other hand, simpler systems are easy to use but lack accuracy and efficiency.

There is a need for a system that combines the advantages of both approaches while minimizing their limitations. The system should be cost-effective, user-friendly, and capable of providing real-time updates without requiring specialized hardware. It should also reduce dependency on manual processes and improve overall efficiency.

3.7 Limitations in Cost and Scalability

Cost is a major factor that affects the adoption of inventory management systems. Advanced systems with better features are often expensive and require additional investment in hardware and software. Small businesses may find it difficult to afford such systems. Moreover, scalability is another issue, as many systems are not designed to handle increasing data and user requirements efficiently.

3.8 Limitations in Usability and Accessibility

Usability is a critical factor for the success of any system. Many existing inventory systems are not user-friendly and require technical knowledge to operate. This creates a barrier for small business owners who may not have a technical background. Complex interfaces and difficult navigation reduce user efficiency and increase the chances of errors.

Chapter 4

Problem Statement and Objectives

4.1 Problem Statement and Objectives

Inventory management is an essential part of retail and business operations, yet many small businesses still rely on manual methods such as registers and spreadsheets. These methods are inefficient, time-consuming, and highly prone to human errors. Mistakes in stock records can lead to inaccurate inventory data, causing issues like overstocking, stock shortages, and financial losses.

Existing digital inventory systems provide some level of automation, but they are often complex, expensive, and require technical knowledge to operate. Many systems depend on dedicated hardware such as barcode scanners and stable internet connectivity, which increases the overall cost and limits their accessibility for small businesses.

Therefore, there is a need for a simple, cost-effective, and efficient inventory management system that reduces manual effort, improves accuracy, and provides real-time updates without requiring specialized hardware. The proposed system addresses these challenges by using a webcam-based barcode scanning approach integrated with a lightweight and user-friendly system.

4.2 Problem Statement

The current inventory management practices used in small retail businesses are inefficient and prone to errors due to manual data handling and lack of automation. Existing digital solutions are either too expensive or too complex to implement and maintain. Additionally, the dependency on specialized hardware such as barcode scanners increases the cost and reduces accessibility. As a result, small businesses face difficulties in maintaining accurate stock records, tracking inventory in real time, and making effective decisions. This creates a need for an inventory management system that is affordable, easy to use, and capable of providing accurate and real-time stock updates without requiring advanced technical knowledge or expensive hardware.

4.3 Objectives

- To develop a simple and efficient inventory management system for small businesses
- To reduce human errors in inventory tracking and data entry
- To provide real-time updates of stock using barcode scanning
- To eliminate the need for dedicated barcode scanners by using a webcam
- To design a user-friendly interface for easy operation
- To maintain accurate and organized inventory records

- To improve efficiency and reduce time required for inventory operations
- To create a cost-effective solution suitable for small-scale businesses

4.4 Scope of the System

The scope of the proposed system is focused on developing a basic and efficient inventory management solution suitable for small retail businesses. The system is designed to handle essential operations such as product addition, barcode scanning, stock updating, and record maintenance. It provides real-time tracking of inventory using a webcam-based barcode scanning approach, ensuring accuracy and efficiency.

The system operates in a single-user environment and uses a local database for storing product and transaction data. It does not include advanced features such as cloud integration, multi-user access, or predictive analytics in its current version. However, the system is designed to be scalable, allowing future enhancements to be added as per requirements.

The scope also includes evaluating system performance based on parameters such as accuracy, response time, and ease of use. This ensures that the system meets the requirements of small businesses effectively.

4.5 Expected Outcomes

The proposed system is expected to significantly improve the efficiency and accuracy of inventory management processes. By automating stock updates through barcode scanning, the system reduces manual effort and minimizes human errors. This leads to more reliable inventory records and better decision-making.

The use of a webcam instead of a dedicated barcode scanner makes the system cost-effective and accessible to small businesses. The user-friendly interface ensures ease of operation, even for users with limited technical knowledge. Additionally, the system provides real-time updates, enabling users to track inventory effectively and respond quickly to changes in stock levels.

Overall, the system is expected to enhance productivity, reduce operational time, and provide a practical solution for inventory management in small-scale business environments.

Chapter 5

Proposed System

5.1 Proposed System

The proposed system is a barcode-based inventory management solution designed to improve the efficiency, accuracy, and reliability of stock management in small retail businesses. The system addresses the major limitations of existing manual and digital systems by introducing automation, real-time processing, and a cost-effective approach. It eliminates the dependency on traditional methods such as handwritten records and spreadsheets, which are prone to human errors and inefficiencies.

The system utilizes a webcam to capture barcode images and applies image processing techniques to detect and decode the barcode. This eliminates the need for dedicated barcode scanning hardware, thereby reducing the overall cost of implementation. The integration of modern technologies such as Python, Flask, OpenCV, pyzbar, and MySQL ensures that the system is lightweight, efficient, and easy to deploy.

The proposed system is designed with a focus on usability and accessibility. It provides a simple and intuitive interface that allows users to perform various inventory operations such as adding products, updating stock, and viewing inventory records. By automating these processes, the system reduces manual effort, minimizes errors, and improves overall productivity.

5.2 System Analysis and Framework

The system analysis focuses on understanding the requirements and designing a framework that ensures smooth interaction between different components. The proposed system follows a modular architecture consisting of input, processing, and output stages.

5.2.1 Input Analysis

The primary input to the system is the barcode captured using a webcam. The system continuously captures frames from the camera and processes them to identify barcodes. The quality of input plays a crucial role in the performance of the system. Factors such as lighting conditions, camera resolution, and positioning of the barcode affect the accuracy of detection.

To ensure reliable input, the system includes basic preprocessing techniques such as image enhancement and noise reduction. These techniques improve the clarity of the captured image and increase the chances of accurate barcode detection. Proper input handling ensures that the system performs efficiently under different conditions.

5.2.2 Processing Mechanism

The processing stage is the core component of the system. It involves detecting and decoding the barcode from the captured image. The system uses OpenCV for image processing and applies techniques such as grayscale conversion, thresholding, and edge detection to identify the barcode region.Once the barcode is detected, the pyzbar library is used to decode the information stored in the barcode. The decoded data is then sent to the backend, where it is processed further. This stage ensures real-time processing and quick response, which is essential for efficient inventory management.

5.3 Design Details

The design of the proposed system focuses on simplicity, modularity, and efficiency. The system architecture is divided into frontend and backend components, ensuring better organization and scalability.

5.3.1 System Architecture

The system architecture consists of three layers: the presentation layer, the application layer, and the data layer. The presentation layer includes the user interface, which allows users to interact with the system. The application layer handles processing, business logic, and communication between components.

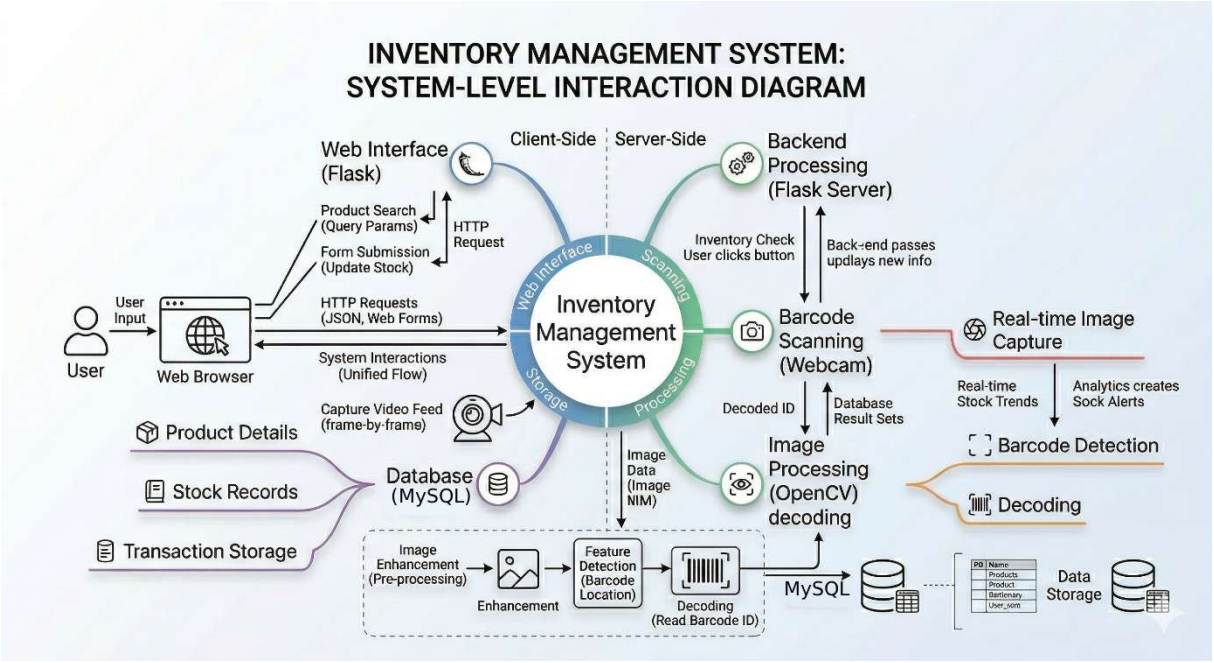


Figure 1 Architecture and Workflow of Web-Based Inventory Management System

5.3.2 Database Design

The database is implemented using MySQL, which is lightweight and efficient for small-scale applications. The database contains tables for storing product information and transaction records. Each product is identified by a unique barcode, which acts as a primary key. The database is designed to support fast data retrieval and efficient storage. Proper indexing and data organization ensure that queries are executed quickly. This improves system performance and ensures that inventory data is always up to date.

5.3.3 User Interface Design

The user interface is designed to be simple, clean, and easy to use. It allows users to perform operations such as adding products, scanning barcodes, updating stock, and viewing inventory details. The interface is designed in such a way that even users with minimal technical knowledge can operate the system efficiently.The layout is organized and intuitive, ensuring smooth navigation. Proper feedback mechanisms are included to inform users about successful operations or errors, improving the overall user experience.

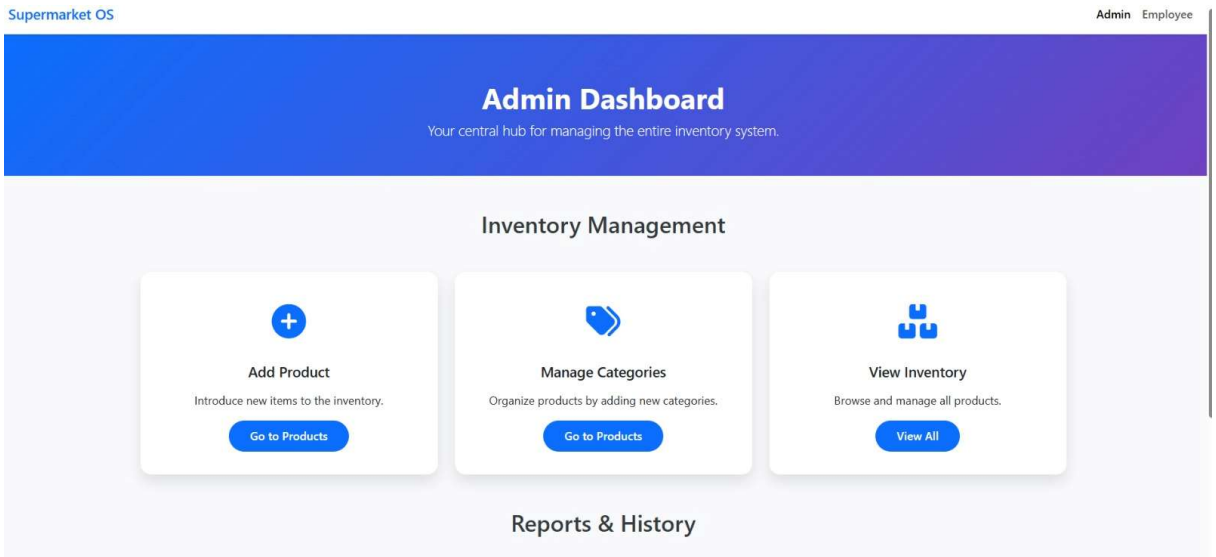


Figure 2 Admin dashboard

← Back to Products

Stock Movement History

Complete audit trail of all inventory operations (BUY, SELL, RETURN)

1	9	1	11
Buy Operations	Sell Operations	Return Operations	Total Operations
10 items	30 items	4 items	All time

Product SKU	Product Name	Category	Operation	Quantity	Timestamp
10	Specs	Skin Care	SELL	1	2026-02-13 08:33:34
10	Specs	Skin Care	SELL	2	2026-02-13 06:39:09
35	Mobile	Electronics	RETURN	4	2026-02-13 06:25:09
35	Mobile	Electronics	SELL	10	2026-02-13 06:24:56
2345	Sun Screen	Skin Care	SELL	2	2026-02-13 06:24:23
2345	Sun Screen	Skin Care	BUY	10	2026-02-13 06:24:13
2345	Sun Screen	Skin Care	SELL	1	2026-02-12 09:06:40
2345	Sun Screen	Skin Care	SELL	3	2026-02-12 08:29:36
2345	Sun Screen	Skin Care	SELL	5	2026-02-12 07:19:46

Figure 3 Stock Movement History

Supermarket OS

Admin Employee

Employee Dashboard

Tools for daily supermarket operations.

Scan Product

Quickly scan items for selling, restocking, or returns.

Open Scanner

Process a Return

Handle customer returns efficiently.

Go to Returns

Figure 4 Employee Dashboard

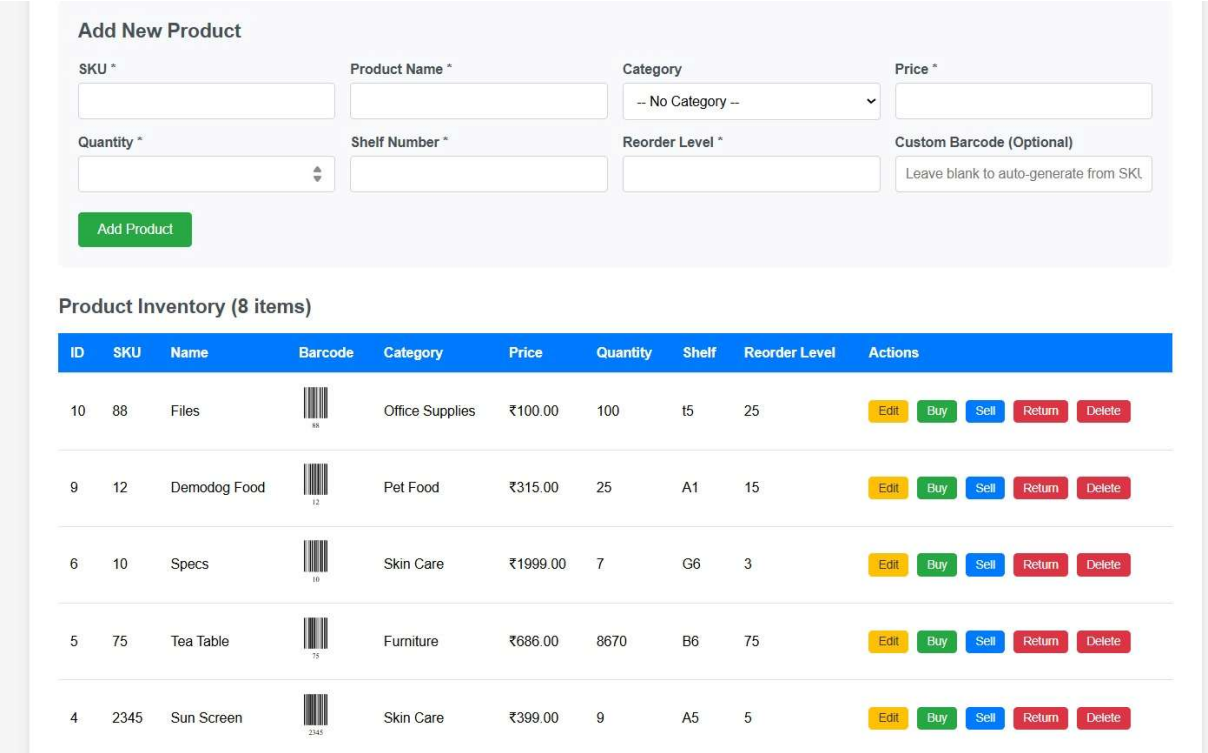


Figure 5 Product Inventory

5.4 Methodology

The system starts by initializing the webcam to capture real-time images. The captured frames are processed using image processing techniques to detect barcodes. Once a barcode is detected, it is decoded to extract product information. The system then checks the database for the corresponding product. If the product exists, its stock is updated. If not, the user is prompted to add a new product. The updated information is stored in the database and displayed on the user interface.

This methodology ensures real-time processing, accuracy, and efficiency. It also reduces manual intervention, making the system more reliable.

5.4.1Algorithm

The algorithm begins with initializing the system and activating the webcam. The system continuously captures image frames and processes each frame to detect barcodes. When a barcode is detected, it is decoded to extract product information. The system then searches the database for the product. If the product is found, the stock is updated; otherwise, a new product entry is created. The updated data is saved in the database and displayed to the user. The process continues until the system is stopped.

This algorithm ensures continuous operation, real-time updates, and efficient inventory management.

5.5 Advanced Features and Functional Enhancements

5.5.1 Point of Sale (POS) & Billing Logic

The system implements a multi-product cart-based selling interface. Users can add items via camera-based barcode scanning or manual override for damaged labels. The billing engine follows a strict financial flow:

$$Subtotal - Discount = Taxable\ Amount$$

$$Taxable\ Amount + GST = Final\ Total$$

5.5.2 Inventory Intelligence & Analytics

- An integrated analytics engine processes stock movement logs to provide:
- Revenue Tracking: Daily, monthly, and yearly financial summaries.
- Product Performance: Identification of top-selling items and highest revenue generators.
- Inventory Health: Detection of "Dead Stock" (items with no movement) to optimize shelf space.
- Profitability Analysis: Accurate profit calculation based on the dual pricing model (*Sell Price - Buy Price*).

5.5.3 Operational Resilience

To ensure zero downtime in fast-paced retail environments, the system includes a manual barcode fallback option. This ensures that even if a barcode is physically damaged or the webcam fails to decode, the transaction can be completed using manual Stock Keeping Unit (SKU) entry.

5.6 Advantages of Proposed System

- Reduces human errors by automating data entry
- Ensures accurate stock updates in real time
- Cost-effective solution (no need for dedicated barcode scanners, uses webcam instead)
- Improves overall efficiency by reducing time required for inventory operations
- Provides a user-friendly interface suitable for non-technical users
- Maintains proper transaction records for tracking inventory movement
- Helps in analyzing business performance through recorded data
- Flexible system that supports future enhancements
- Scalable for features like cloud integration and multi-user access
- Reliable and efficient solution for small business inventory management

Chapter 6

Experimental Setup

6.1 Experimental Setup

The experimental setup of the proposed barcode-based inventory management system is designed to evaluate its performance, accuracy, and usability in a controlled environment that closely resembles real-world conditions. The setup involves configuring the necessary hardware and software components, preparing the database, and defining test scenarios to validate system functionality. The aim of this setup is to ensure that the system operates efficiently and meets the requirements of small retail businesses.

The system is tested by scanning barcodes using a webcam under different conditions, such as varying lighting environments and different barcode orientations. Multiple test cases are considered, including adding new products, updating stock quantities, retrieving product information, and handling invalid or missing barcodes. This approach ensures that the system is robust and capable of handling real-world challenges effectively.

The experimental setup also focuses on evaluating the system's response time and accuracy during continuous operation. The system is executed multiple times to observe consistency in performance and to identify any potential issues that may arise during long-term usage.

6.2 Database and Input Details

The database used in the proposed system is implemented using MySQL, which is lightweight, efficient, and suitable for small-scale applications. The database consists of structured tables designed to store product details and transaction records. Each product entry includes attributes such as product ID, product name, barcode number, quantity, and price. The barcode serves as a unique identifier, ensuring that each product can be accessed and updated efficiently.

The database is designed to support fast data retrieval and ensure data consistency. Proper indexing techniques are used to improve query performance, allowing the system to handle multiple operations efficiently. The database also maintains transaction records, which help in tracking inventory movement and analyzing system performance over time.

The input to the system is provided through a webcam, which captures real-time images of product barcodes. These images are processed using image processing techniques to detect and decode the barcode information. The system is tested using a dataset of approximately 50 to 100 sample products, each assigned a unique barcode. This dataset is used to evaluate the system's ability to handle multiple entries and perform accurate updates.

The input handling mechanism includes validation checks to ensure that only valid barcode data is processed. In cases where the barcode is not detected or does not exist in the database, the system prompts the user to add the product manually. This ensures data integrity and prevents incorrect entries in the system.

6.3 Performance Evaluation Parameters

- Accuracy: Measures the system's ability to correctly detect and decode barcodes, ensuring reliable inventory records and minimizing errors
- Response Time: Refers to the time taken to process a barcode and update the database, ensuring fast and efficient operations
- Reliability: Evaluates consistency of system performance under different conditions such as lighting, camera angles, and repeated usage
- Usability: Determines how easily users can interact with the system through a simple and intuitive interface without technical expertise
- Efficiency: Compares the time required for operations with traditional manual methods, improving overall productivity
- Overall Evaluation: These parameters together help validate the effectiveness, reliability, and practicality of the proposed system

6.4 Software and Hardware Setup

- Python Integration: Used as the primary programming language due to its simplicity, flexibility, and extensive library support.
- Flask Web Framework: Utilized for handling backend operations and creating the intuitive user interface.
- OpenCV Library: Implemented for core image processing tasks to identify barcode regions within captured frames.
- Pyzbar Library: Specifically used for the high-speed decoding of barcode information from processed images.
- MySQL Database: Employed as the lightweight database management system for efficient data storage and seamless integration.
- Webcam-Based Input: Uses a standard webcam to capture real-time images of barcodes, eliminating the need for specialized hardware.
- Hardware Accessibility: Designed to run on a standard computer or laptop, making it cost-effective and accessible for small businesses.

- Minimum Requirements: Operates efficiently on devices with a basic processor, sufficient RAM, and a webcam with adequate resolution.
- Standardized Testing: Validated on common laptop configurations to ensure stable performance on widely available hardware.
- System Deployment: Engineered for easy installation without complex configurations, ensuring suitability for users with limited technical knowledge.
- Operational Readiness: The integrated setup ensures the system is practical, scalable, and fully prepared for real-world implementation.

6.5 Testing and Validation

The testing phase is an essential part of the experimental setup, as it ensures that the system functions correctly under different scenarios. The system is tested using multiple barcode samples to verify its ability to detect and decode barcodes accurately. Various test cases are considered, including valid barcode detection, invalid barcode handling, and database update operations.

Validation is performed by comparing the system output with expected results. The accuracy of barcode detection is measured by calculating the percentage of correctly identified barcodes. The system is also tested for response time and reliability by performing repeated operations.

The results of testing indicate that the system performs efficiently and provides accurate inventory updates. Minor variations in performance are observed under poor lighting conditions, but overall system reliability remains high. This validation confirms that the proposed system meets the requirements of small retail businesses.

Chapter 7
Results and Discussion

7.1 Results and Discussion

The results of the proposed barcode-based inventory management system demonstrate its effectiveness in improving accuracy, efficiency, and ease of use in inventory operations. The system was tested under various conditions to evaluate its performance in real-world scenarios. The results obtained from testing indicate that the system successfully detects and decodes barcodes using a webcam and updates inventory records in real time.

The system was able to handle multiple operations such as adding new products, updating stock levels, and retrieving product details with high accuracy. The integration of image processing techniques with database management ensured that inventory data was processed efficiently. The results also show that the system reduces manual effort and minimizes human errors, making it a reliable solution for small retail businesses.

7.2 Performance Results

The performance of the system is evaluated based on parameters such as accuracy, response time, and reliability. The system achieved high accuracy in barcode detection under normal lighting conditions, with most barcodes being correctly identified and processed. The response time for updating inventory records was observed to be low, ensuring quick and efficient operation.The system also demonstrated consistent performance during repeated operations, indicating high reliability. Minor variations in performance were observed under poor lighting conditions; however, the overall accuracy remained acceptable. These results confirm that the system is capable of performing efficiently in real-world environments.

7.3 Discussion of Results

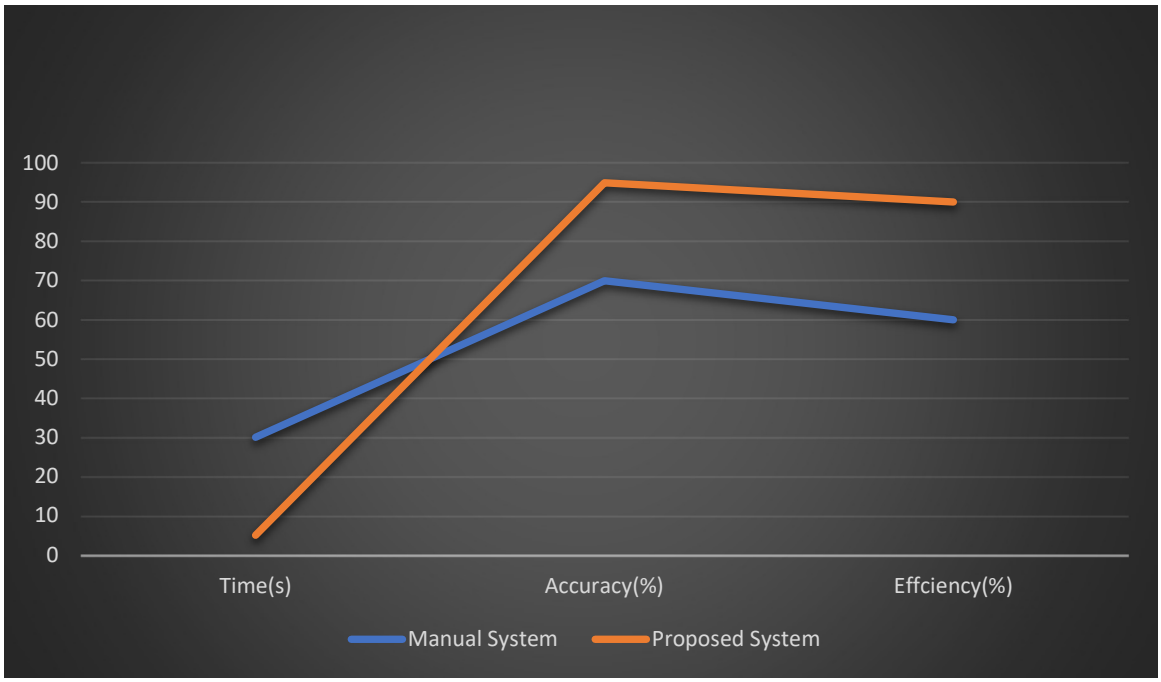
The results indicate that the proposed system effectively addresses the limitations of traditional and existing inventory management systems. By automating barcode scanning and inventory updates, the system reduces dependency on manual processes and minimizes errors. The use of a webcam instead of a dedicated barcode scanner significantly reduces the cost of implementation while maintaining functionality.

The system also provides real-time updates, which improves decision-making and enhances overall efficiency. The user-friendly interface ensures that the system can be easily operated by users without technical expertise. The discussion highlights that the proposed system is a practical and scalable solution for small businesses.

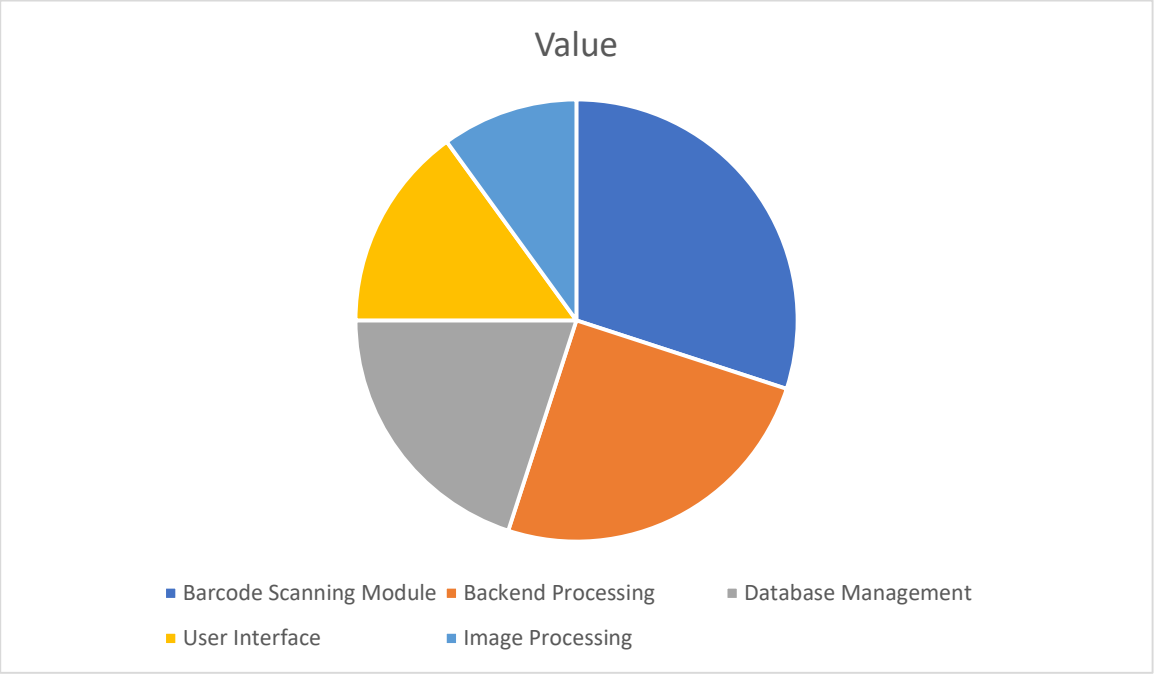
However, certain limitations were observed during testing. The performance of the system is affected by lighting conditions and camera quality, which can impact barcode detection accuracy. Despite these limitations, the system performs reliably under normal conditions and meets the requirements of small-scale inventory management.

7.4 Graphical Representation and Analysis

The results obtained from the system can be represented using graphs and charts to provide a better understanding of performance. Parameters such as accuracy, response time, and efficiency can be visualized using bar charts, line graphs, or pie charts. These graphical representations help in analyzing system performance and identifying areas for improvement. For example, a bar graph can be used to compare the accuracy of barcode detection under different lighting conditions, while a pie chart can represent the distribution of successful and failed scans.



Graph 1 Performance Comparison of Proposed System Parameters



Graph 2 Distribution of System Components in Proposed Inventory System

Chapter 8

Conclusion and Future Work

8.1 Conclusion and Future Work

The proposed barcode-based inventory management system successfully addresses the limitations of traditional and existing inventory systems by providing an efficient, accurate, and cost-effective solution. Overall, the system achieves its objectives by providing a practical and scalable solution for inventory management. It improves productivity, reduces operational time, and ensures accurate maintenance of inventory records. The results obtained from testing and validation confirm that the system performs efficiently under normal conditions and meets the requirements of small-scale business environments.

8.2 Conclusion

The development of the proposed system highlights the importance of automation in improving business operations. By replacing manual processes with a technology-driven approach, the system enhances accuracy and efficiency in inventory management. The use of modern tools such as Python, Flask, OpenCV, and MySQL ensures that the system is lightweight, reliable, and easy to implement.

The system successfully demonstrates how image processing and barcode technology can be integrated to create a cost-effective solution. It provides real-time inventory tracking and maintains proper records, which helps in better management and decision-making. The project proves that advanced inventory management solutions can be developed using simple and affordable technologies.

The project successfully transitions from a basic inventory tool to a full-scale retail management platform. By integrating POS capabilities with a backend intelligence layer, the system provides small business owners with the same level of data-driven insight typically reserved for large-scale enterprise resource planning (ERP) software. It proves that a cost-effective stack (Python, Flask, and OpenCV) can handle complex financial logic and real-time analytics.

8.3 Future Work

Although the proposed system performs efficiently, there are several areas where it can be further improved. Future enhancements can include the integration of cloud-based storage, which would allow remote access and multi-user functionality. This would make the system more scalable and suitable for larger businesses. The system can also be extended by developing a mobile application, enabling users to manage inventory using smartphones. Advanced features such as data analytics and demand prediction using artificial intelligence can be added to improve decision-making. Additionally, improving barcode detection under low-light conditions and enhancing image processing algorithms can further increase system accuracy.

Beyond these technical refinements, the platform can be professionalized through several functional upgrades. The system can be expanded to include automated receipt generation by integrating thermal printer APIs for physical billing, alongside a dedicated Customer Relationship Management (CRM) layer to track purchase history and manage loyalty programs. Financial flexibility can be improved by incorporating payment gateway integrations, specifically supporting UPI and digital wallet payments directly within the POS workflow. Furthermore, the existing analytics data provides a foundation for predictive stocking; by applying Machine Learning algorithms to historical sales trends, the system could forecast future demand, helping small business owners optimize their inventory levels and significantly reduce waste.

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